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Mentor Hints for Week 6 Quiz Part II

Heng Ye · Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ) · Edited (/learn/analytics-excel/promises/3e0094631daa57d9105380de17413dc17)

Click to download week 6 Excel workbook (<https://www.coursera.org/learn/analytics-excel/supplement/5Tld7>)

Common Mistakes

When you use the test dataset, don't forget to use the outcome for the test dataset too. [Outcome is the column with 1 and 0 that shows whether the cardholder defaulted or not]

Question 1 What is the AUC of the Eggertopia Scores on the Training Set?

Copy paste the Eggertopia Credit Scores (1-200) into the AUC calculator tab at A23 and read the value from the tab.

Question 2 What is the optimum threshold on the training set to minimize the average cost per test?

Look for the threshold level at the minimum cost

Question 3 What is the average cost per test at the training set optimum threshold?

The average cost per test at the training set optimum threshold is shown in the minimum cost cell located at the upper right corner.

Question 4 What is the AUC of the Eggertopia scores on the Test Set?

Copy paste the Eggertopia Credit Scores (201-400) into the AUC calculator tab at A23 and read the value from the tab.

Question 5 Using the same threshold as used on the training set, what is the cost per event of the Eggertopia scores on the Test Set?

At the threshold of Question 2, what is the cost for the test set?

Question 6 What is the maximum, (break-even) price that the bank could theoretically pay per score for Eggertopia Data, if it did not have your model and its data – but simply issued cards to everyone who applied?

Hint: In other words, what is the difference in cost per event to the bank at a 25% default rate, and at the default rate (False Negative Rate) and False Positive rate of the Eggertopia scores at their optimum threshold?*

\$1250 - the answer in Question 5

Question 7 The True Positive rate?

Shown in the row 12 and under the threshold identified in Question 2

Question 8 What is its Positive Predictive Value?

Positive Predictive Value (PPV) = e/c (Please refer to Binary Classification workbook)

Question 9 What is the Negative Predictive Value?

Negative Predictive Value (NPV) = h/d (Please refer to Binary Classification workbook)

Question 10 What is the “test incidence” of Eggertopia’s classification, on the test set, at the threshold from the training set? In other words, what percentage of applicants does Eggertopia classify Positive as “defaulters” and what percentage Negative as “non-defaulters” on the Test Set?*

$TP + FP / \text{total population} = \text{number of defaulters; non-defaulters} = \text{total population} - \text{defaulters}$

Question 11 What is the maximum, or break-even, price that the bank could pay per score for Eggertopia, given that it already has your model and data?

Value per score for Eggertopia - Value per score for your model

Why divide by “-1”?

The purpose of “-1” is used to make direction of the score right, namely lower score to higher chance of default and higher score to lower chance of default. You won't need it if your model has it in the right direction.

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Using the same threshold as used on the training set, what is the cost per event of the Eggertopia scores on the Test Set?*

Is there an error on this answer? I have checked and double checked my answer and am pretty sure that the correct option wasn't available. Hence I picked an option that was \$1 away. As my other answers were correct, I think perhaps the correct option was not shown?

Thx, K

https://www.coursera.org/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/IT_VmpAgEeWSvw7ngxeQVw

1/10

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Heng Ye · Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/sTzh8li0EeW_NAqM69GjBQ)
(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)
I agree with you, the options are updated now.

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Rob Nicholas · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/SzrDyovXEeWD3RImE8PCdQ)
(/learn/analytics-excel/profiles/ae1346f82d1a8de6bb800d9a2f514539)

Hi Heng, 4 days after you updating this, I am looking at the same issue...I found an answer that was \$1.00 away from what shows in my Spreadsheet, and it's marked wrong. Wonder what I'm doing wrong...

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Heng Ye · Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/ZhijmYzYEeWrDQr9ZhnP1w)
(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)
What is the range of the standardized score? Is it still from -1.47 to 3.29?

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Angie · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/3LSCpYzYEeWD3RImE8PCdQ)
(/learn/analytics-excel/profiles/5d5e809eca29b2322e7b7050e9b922f0)

Hi Heng, I have that range, Heng, but the answer option is still not there as of 7pm Pacific time, 11/16/15. I was marked wrong as I chose the closest option \$839. So frustrated!!!

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Satish Chandra Reddy, N · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/zVG8KYz1EeWD3RImE8PCdQ)
(/learn/analytics-excel/profiles/7014d2cb1025168953a98e241d884c3b9)

Hi Angie, I see \$833, 838, 823,803 - 17 Nov 2015, afternoon

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Angie · 24 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/u0kgdl4UEeWJvQrEo78pXQ)
(/learn/analytics-excel/profiles/5d5e809eca29b2322e7b7050e9b922f0)

That's strange. I didn't see the \$838 option. I chose \$839 and was marked wrong.

▲ 0 Upvote



Aaron Mayzes · 15 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/xE2y4ZWKeeWb8A4PYD_NBw)
(/learn/analytics-excel/profiles/93a5a346a6d917bc09b7a26894d349d8)

I also did not have the \$838 option.

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Ricardo Torres · 14 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/GtgRYIfgEeWDahl4As6YoQ/comments/Fs3zZX1EeWTngqUjd-yuQ) · Edited
(/learn/analytics-excel/profiles/026ad4c01925729f38184ab240d13)

Hi Heng, I do not have the 838 as an option but when taking the closest one the system is marking it as wrong. can you please do let me know your comments on this?

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Update: Never mind just retook the test and the correct options appeared

cheers

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SD

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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Jennifer Xu · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65h11lbwj7Q)
(/learn/analytics-excel/profiles/efb5a86e8321cc0594d4c38c98614aa7)

Hi Heng, I did not have the 838 as an option but when taking the closest one the system is marking it as wrong. can you please do let me know your comments on this?

\$1250 as the minimum cost, and the theshold is wayyyy far away from the answers. Something is not right. Please help!

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Damien Papworth · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65h11lbwj7Q/comments/AtfAMYIUeEwJvQrEo78pXQ)
(/learn/analytics-excel/profiles/a09f94f04593086de2a7fa7077722f8e)

My guess is that your spreadsheet is still dividing the Eggertopia score by -1 (cell C20 on the AUC calculator spreadsheet).

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Jennifer Xu · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65h11lbwj7Q/comments/W6ZRMYYEeWwEwr2ChLK8Q)
(/learn/analytics-excel/profiles/efb5a86e8321cc0594d4c38c98614aa7)

You are right. How do I know if I should use 1 or -1 ? Now, I have the same number on AUC and "area under rectangle". I know the answer makes sense now. Thank you very much!

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Heng Ye · Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65h11lbwj7Q/comments/23rFdoIgEeWD3RImE8PCdQ)
(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

The -1 is divided so that higher score means better chance of non-default. But if you are designing your score in

excel/profiles/11e005485fdae37d9fb580de1415dc1) the right direction, you don't need that -1.

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Scott Young · 15 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65hl1lbwj7Q/comments/F1A0YZUHEeWWpRIGHRsuuw) (/learn/analytics-excel/profiles/8422e7ac9b1e01231933126c4991d6ad) You be using I201-I403 in Column A and H201 to H403 in Column E?

According to the hint "Copy paste the Eggertopia Credit Scores (201-400) "

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John Brew · 15 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65hl1lbwj7Q/comments/G_sKNZUqEeWtngq0jd-yuQ) (/learn/analytics-excel/profiles/827314bb2ff17aa79121452e3d8f8364) I had the same problem and did not figure it out until I checked here, thanks.

▲ 0 Upvote



Pranay Kumar Kawalkar · 15 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65hl1lbwj7Q/comments/8QuzaZUXeEWPmAru8qzZow) (/learn/analytics-excel/profiles/0c4fc7ef95cc11967fdf6ee8ec426863) I did the same mistake of dividing by -1. Thanks guys.

▲ 0 Upvote



David Gerrard · 9 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65hl1lbwj7Q/comments/XcXw75oHEeWuURKvSnbBUQ) (/learn/analytics-excel/profiles/e820e2bd5d0c96c6bd18bf3f8636810) If you are designing your score in the right direction, you don't need that -1."

I'm afraid this doesn't make any sense to me. What does it mean to "design your score in the right direction"?

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John Brew · 9 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/kVHVR4ILEeW65hl1lbwj7Q/comments/2-ymmj09EeWy0Q7ABZMsnQ) (/learn/analytics-excel/profiles/827314bb2ff17aa79121452e3d8f8364) I am starting a new thread called AUC Calculator Visualized (https://www.coursera.org/learn/analytics-excel/module/5m5Am/discussions/zt9unZo9EeWNbBlwwhtGwQ) that may help.

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SD

Reply

(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Reply

DP

Damien Papworth · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/q8EQSolVEeWrDQr9ZhnP1w)

(/learn/analytics-excel/profiles/a09f94f04593086de2a7fa7677722f8e) Given that Q1 gives the bound for the Eggertopia scores, I've found that normalising the scores to that range will give a different threshold than just using the scores as is. I thought I'd point this out as it took me another attempt to correctly get the answers for all questions.

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/q8EQSolVEeWrDQr9ZhnP1w/comments/k0LRlllIEeWwEwr2ChLK8Q) (/learn/analytics-excel/profiles/31e005485fdae37d9fb580de1415dc1) Yes, that's true. Be sure to utilize the full range.

▲ 0 Upvote



Sandra Humphrey · 18 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/q8EQSolVEeWrDQr9ZhnP1w/comments/78zx-jKbEeW70BJZtLVfGQ) (/learn/analytics-excel/profiles/c65ed68e0d93e0baa1ff2f4b7771b60) Meaning adding to the values to expand the range to fit? I can get a plausible AUC and Cost, but not and expected threshold

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SD

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Reply



Jean Michael Manyari Jimenez · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw) (/learn/analytics-excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9) Help me please! When I change the data to find the AUC of the Eggertopia on the Test Set and change the -1 for 1 like the other one i have a result that It's different from the answers, what I'm doing wrong?

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/AznUDImSEeWwEwr2ChLK8Q) (/learn/analytics-excel/profiles/31e005485fdae37d9fb580de1415dc1) I might have changed the worksheet. Try re-download it.

Copy paste the Eggertopia score to the first column and then you should be able to see the answer.

▲ 0 Upvote

- 

Jean Michael Manyari Jimenez · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/IBXDYImYEeWQmAqk3OqK6Q) · Edited

(/learn/analytics-excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

I already did that and the result for the AUC is 0.53 and it is not in the alternative results

excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/QuCapiImgEeWQmAqk3OqK6Q)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Can you walk me through what you did? Lets figure out what is going on together.

excel/profiles/31e005485fdaae37d9fb580de1415dc1)

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Jean Michael Manyari Jimenez · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/nYo7JYmhEeWv8BKLWolBUQ)

(/learn/analytics-excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

Well first of all I copied from 7-3 Data - Training & Test Sets the Data Test Set (201 - 400) from Eggertopia Credit Score Card. After that, I placed all the information in the first column in 7-1 AUC Calculator to calculate the AUC and change the -1 to 1 and I obtain 0.53. What I'am doing wrong?

excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/Ds6sllmmEeWQmAqk3OqK6Q)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Use the Training set from 1 - 200, not the test set

excel/profiles/31e005485fdaae37d9fb580de1415dc1)

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- 

Jean Michael Manyari Jimenez · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/yySlKYmqEeW65h11bwj7Q)

(/learn/analytics-excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

But in the test told us had we have to use the both

excel/profiles/cd7ad6429a6b0e552d8b32d885ef6dc9)

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/gCINqozHEeWH0w7fQkp2-w)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

see the problem, Don't forget to copy paste the "outcome" too. I get the your result when I didn't do that.

excel/profiles/31e005485fdaae37d9fb580de1415dc1)

▲ 2 Upvote
- 

Gary Chan · 25 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/jzOsA42gEeWrDQr9ZhnP1w)

(/learn/analytics-excel/profiles/111be132498ea2accc136830a76f3d4e)

What is the "outcome". May I know it is in which column? Thanks

excel/profiles/111be132498ea2accc136830a76f3d4e)

▲ 0 Upvote
- JW

James Wisnieski · 24 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/euqGsl4MEeWfyQ7wUMeYyw)

(/learn/analytics-excel/profiles/d9beaf185be0354f8be0f12c251cc3c6)

Gah, made the same mistake as Jean! Gary: the outcome is the 1 or 0 column that gives the outcome of default or not, its column H on sheet 7-3.

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- 

Gary Chan · 24 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/ICDC4I5ZEeWfyQ7wUMeYyw) · Edited

(/learn/analytics-excel/profiles/111be132498ea2accc136830a76f3d4e)

Hi James, Yes I saw the outcome in column H but I though it is exactly same as column E in sheet 7-1 AUC calculator. Where should we copy and paste the column H to 7-1 AUC calculator?

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- 

Gary Chan · 24 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/52FOA45bEeWv8BKLWolBUQ) · Edited

(/learn/analytics-excel/profiles/111be132498ea2accc136830a76f3d4e)

on James, you mean copy the outcome for test set from sheet 7-3 to 7-1 AUC calculator? Yes, I forgot to do it. Now I see what did I go wrong in Quiz 1 also.

▲ 0 Upvote
- JW

Jamie Walsh · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/7trp_VY7mEeW65h11bwj7Q)

(/learn/analytics-excel/profiles/24e1fe8d28b6bf2adde9c41c86d3221)

Yes, that's correct Gary. I just tried it and it worked, for those of you who have doubts and is reading this for the first time. I was just like Gary and am only just realizing that this was my problem in Quiz 1. I had an excellent model that I only just realized. I thought I was just stupid before and couldn't figure it out (which is still true, considering I didn't think to figure out all the fields in the AUC Calculator) ;(

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Adeleke Adetola · 17 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/P3p7UomHEeWJ8Q5UvS9Vkw/comments/4XmwwZOfEeWuoA7BxrrLbQ)

(/learn/analytics-excel/profiles/a07855645bdecbf8a673fca5b72274493)

Hi, when I put in the Eggertopia test set with the corresponding default values in the AUC excel spreadsheet, I got the same AUC value which I got in the training set...i.e 0.85 again! which I know is very wrong. plsss help!

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SD

(/learn/analytics-excel/profiles/f34069ce8df6de7dbefbedfb7e760d9f)

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RB

Robert Bunting · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/mhE1xluEeWZfg6Bj_UARQ)

I am having significant difficultv. I have reviewed the week 2 material and the innumerable discussion posts. and I can only

(/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/mhE1xluieEWZfg6Bj_UARQ/comments/Wl1sdYzEEeWfyQ7wUMeYyw) conclude this week was an absolute mess. Will you please clearly and completely state how we are to derive the true positive rate, the positive predictive value, and the negative predictive value from the Excel spreadsheet? I feel like I missed not one step in the lecture but many.

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/mhE1xluieEWZfg6Bj_UARQ/comments/Wl1sdYzEEeWfyQ7wUMeYyw) (/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

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Review Week 2 content

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SD

Reply

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Angie · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFJozbEeWD3RImE8PCdQ)

A Q7-10: The True Positive rate? What is its Positive Predictive Value? What is the Negative Predictive Value? What is the “test incidence” of Eggertopia’s classification on the test set, at the threshold from the training set? In other words, what percentage of applicants does Eggertopia classify Positive as “defaulters” and what percentage Negative as “non-defaulters” on the Test Set?

Threshold for Scores over range [3.5, -3.5] this row	
TP this row [sum in column from 23:222]	
y = TP rates - this row	
Total False Positives (FPs) at threshold	
x = FP rates - this row	
Area of rectangle ((y1+y2)/2)*(x2-x1) this row	
FN this row - total Condition 1 - TP	
Cost Per Event = (FN*cost per FN)+(FP*Cost per FP)*Total Events	

I've tried the training Eggertopia set:

- 1) Identify the threshold at the minimum cost per event
- 2) Plug the TP (2nd row), Total FP (4th row), FN (7th row) from the table above into the Confusion matrix
- 3) Calculate the answers for Q7-9, and got all wrong. Why????

Q10: Defaulters test incidence = TP/(TP+FN) . Why was my answer marked wrong???

Please help! Thank you!

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Heng Ye Teaching Staff · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFJozbEeWD3RImE8PCdQ/comments/PhRXozpEeWZfg6Bj_UARQ) (/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

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Satish Chandra Reddy, N · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFJozbEeWD3RImE8PCdQ/comments/vKUBRIzzEeWZfg6Bj_UARQ) (/learn/analytics-excel/profiles/014d2cb1025168953a98e241d884c3b9)

Final Project 2, Q3 : What is the average cost per test at the training set optimum threshold i copied 1-200 from Eggertopia Credit Scores and pasted in 7-1 AUC calculator (col A), threshold -1 : got \$1,250 /0.85

I haven't got answer ?

▲ 0 Upvote



Satish Chandra Reddy, N · a month ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFjozbEeWD3RImE8PCdQ/comments/b9LoZ4z0EeWQMqAqk3OqK6Q)
Hi Heng, the answer is 0.53
excel/profiles/014d2cb1025168953a98e241d884c3b9)
Q5 : 1,238

in both cases i could not get answer correct

▲ 0 Upvote



Angie · 24 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFjozbEeWD3RImE8PCdQ/comments/U9f_0o4VEeWZfg6Bj_UArQ)
Hi Heng, I got the other questions correctly i.e. the threshold and the minimum cost per event should be correct.
excel/profiles/5d5e809eca29b2322e7b7050e9b922f0)
▲ 0 Upvote



Jamie Walsh · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFjozbEeWD3RImE8PCdQ/comments/196bGY71EeWImArmHqeQkQ)
Hi Heng, I got all the answers correct, through calculations, except for Q10. I tried to use Angie's Formula from above but that didn't work. I googled it and read through all the transcripts of the lectures but I don't see a formula for calculating test incidence. can someone please be so kind to provide an example with the formula??

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Duncan Patrick Hodgson · 8 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFjozbEeWD3RImE8PCdQ/comments/8rQH7ppkEeWuURKvSnbbUQ)
From Q10 Quiz 1. I think the test incidence is $e/(a+b)$ in the confusion matrix, hope thats right, where (a+b) is the total number of scores. The definition should be added to the excel spreadsheet in week 2.

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Duncan Patrick Hodgson · 8 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/orsFjozbEeWD3RImE8PCdQ/comments/uiCxj5qUEeWuoA7BXrrLBQ)
Had another look at the test incidence in Question 10 and I was wrong. Question 10 in Quiz part 2 spells it out as $(total\ scores)/total\ scores\ ie\ (a+b)/(a+b)$

sorry about the previous post

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Reply



Prannav Khanna · 25 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/DmJLGyZ_EeW65h11Ibwj7Q)
Hi!
I am trying to figure out how to go about Q6. Please help!

Also, I'd really appreciate if you can share some insights on Q10 too. I couldn't get a correct answer.

Thanks,

Pranav

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Gary Chan · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/DmJLGyZ_EeW65h11Ibwj7Q/comments/yjFYml6ZEeWH0w7fQkp2-w)
I think the Q6 is turning the +1 to -1, which is \$1,250 and using the amount to minus the answer in Q5
excel/profiles/111be132498ea2acec136830a76f3d4e)
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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Reply



Satish Chandra Reddy, N · 25 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/lvZyfY0DEeWD3RImE8PCdQ)
Hi Prannav,
Are you able to figure out Q4, 5, 6, 7, 8, 9? If so could you give a hint - Thanks

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Heng Ye Teaching Staff · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/lvZyfY0DEeWD3RImE8PCdQ/comments/8ndsVI8BEeWUChK_4yGk2Q)
Hi Satish, check the main body.
excel/profiles/31e005485fdaae37d9fb580de1415dc1)
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Satish Chandra Reddy, N · 22 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/lvZyfY0DEeWD3RImE8PCdQ/comments/1b61VIAVEeWmH0w7fQkp2-w)

 excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q) (/learn/analytics-excel/profiles/014d2cb1025168953a98e241d884c3b9)

for Ques 5 i got different results
\$ / Threshold / AUC / copied or not copied Col H (7-3) - Col E (7-1)

738 / 0.10 / 0.84 / COPIED col H from sheet 7-3 and paste it to column E at 7-1 sheet

1,425 / 0.10 / 0.94 / NOT COPIED column H from sheet 7-3 and paste it to column E at 7-1 sheet

why i am not able to get the results ?

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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Crystal Tang · 25 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q)

Hi, Heng,

can you give a little hint on the last question? Question: What is the maximum, or break-even, price that the bank could pay per score for Eggertopia, given that it already has your model and data?
I am a bit stuck here, thanks.

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 Heng Ye Teaching Staff · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/_mV4Ao88EeWerQ55qE4MgQ)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Hi Crystal, are you able to see the hints after submitting your answers?

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 Heng Ye Teaching Staff · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/INQD118CEeWSv7ngxeQVw)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Question 11 is asking you the different in value per score between your own model and Eggertopia.

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M

Mini · 23 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/BJMsPo8UEeWwqwoBxhpijw)

(/learn/analytics-excel/profiles/0ad3d4b7e4531aa497a52af9583b39ec)

For question 11, should the difference be between values from the *test* data but using the respective *threshold* from the *training* data for both my model and Eggertopia?

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 Heng Ye Teaching Staff · 21 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/oRPMM5BUeEw70BJztlVfGQ)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Yes, always use the same threshold identified from training set to test set evaluate the model.

Question 11 What is the maximum, or break-even, price that the bank could pay per score for Eggertopia, given that it already has your model and data?

Value per score for Eggertopia - Value per socre for your model

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M

Mini · 21 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/INhPlpB3EeW70BJztlVfGQ)

(/learn/analytics-excel/profiles/0ad3d4b7e4531aa497a52af9583b39ec)

Thanks Heng!

▲ 0 Upvote

EM

Eric Marzari · 17 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/BocjmjNzEeWKNwpBrKr_Fw)

(/learn/analytics-excel/profiles/e037ba84f980419de627a69ae373c591)

Hi Heng, what if the minimum cost per event from my model is already lower than the one provided by Eggertopia, please ?

Is there an alternate way to price whatever metric (e.g. the AUC, in my case) is better with Eggertopia, instead of the cost per event ?

Thanks

EM

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 Andrew Tyriver · 8 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/5-1AupsHEeW0dA6KeRPXvQ)

(/learn/analytics-excel/profiles/d78aca9e41d383762f99b43ecf44098)

Following question Heng: is it the test sets for both the Eggertopia and my model? Is the value per score the same as cost per event?

So is this the correct formula: Cost per event for Eggertopia test set - Cost per event for my model on the test set (from Quiz 1/Part 1)?

▲ 0 Upvote



Heng Ye Teaching Staff · 4 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Gzff-429EeWwEwr2ChLK8Q/comments/DesH4p3gEeWn4A5xojIPDw)

(/learn/analytics-

Hi Andrew, sorry for late reply.

excel/profiles/03e9431cebc844eae0376748713233c5) and my model?

Yes, *Eggertopia* is just another model for the same dataset.

Q: Is the value per score the same as cost per event?

Value per score is the difference of the cost per event between your model and the default rate (\$1250)

Q: So is this the correct formula: Cost per event for Eggertopia test set - Cost per event for my model on the test set (from Quiz 1/Part 1)?

This is the right formula for the break-even price in Question 11.

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excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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MW

Malwina Wnorowska · 22 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/er4-ujAREeWNOW5YyoNGsw)

(/learn/analytics-

I think there is something really wrong with the scoring system. Most, but not all, of the answers are exactly the same (marked as correct) as the sample answers.

Also, the same answer to a question that was marked as correct at first submission was marked incorrect at 2nd submission.

Very confusing and definitely not helping to understand the material of this course.

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Heng Ye Teaching Staff · 21 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/er4-ujAREeWNOW5YyoNGsw/comments/9FVi_ZBTeeWKNwpBrKr_Fw)

(/learn/analytics-

Thanks for feedback Malwina, I will report to Coursera Help Center for this issue. If you can provide me screenshots as evidence it will help me to make a stronger argument to them.

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excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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Satish Chandra Reddy · 21 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/IT_VmpAgEeWSw7ngxeQVw)

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Hope many of you have done this Part 2 Quiz : i request please post you way of solving the below ques including Heng - Thank you

Question 5 :

Doubt : I am not able to get the answer provided in the quiz . Ques 2 showed 0.1 (as optimum threshold) /\$588/AUC0.77, and rest of the threshold did not show significant change in \$ but change in AUC.

Question 7

Doubt : If the threshold for ques 2 is 0.1 , then is should come under BX12

Question 8 & 9 :

Doubt : e = True positives, c = classification incidence "POS", where do i get this values from ?. Similarly for h and d (from Ques 9)

Question 10 : TP + FP / total population = number of defaulters; non-defaulters = total population - defaulters

Doubt : where do i get this values is it from excel ?

Question 11:

Doubt : your model : does it refer to Part 1 Quiz 1 ?

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Kaitlin Caravello · 18 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/qjLmqZMEeWB8gqILqx0tQ)

(/learn/analytics-

Question 11:
After reviewing the answer, I am still confused on how to calculate the *Value per score* for *Eggertopia*

Given formula: *Value per score for Eggertopia* - *Value per score for your model*

From the AUC tab for the test set, using the training set's minimizing threshold, I was able to calculate the *Value per score for your model*

Can you please explain where the *Value per score for Eggertopia* is coming from, and how to calculate it?

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Syed Ali Mudassar Kazmi · 10 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/qjLmqZMEeWB8gqILqx0tQ/comments/Fi5_zpjUEeWB8gqILqx0tQ)

(/learn/analytics-excel/profiles/7b726d35c2a92bd39c9c90ce6c66f52f)

can you plz tell me how you calculated *Value per score for your model?*

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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AR

Adam Rose · 17 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/qkOdW5PLEeWy0Q7ABZMsnQ)

I need clarification on the proper use of normalizing the data set and how that affects the AUC. When normalizing the Eegertopia data set to closely match the -3.5 to 3.5 range to the extent that you get it close or very close to this set it drastically effects the distribution of the data and the correct Threshold. If you use a range of 0.94 to -2.45 you end up with an threshold of -0.15, but if you keep tinkering to get it to 3.2 to -3.45 then the threshold changes to 0.10. How do I know when I have normalized it enough? There seems to be a lot of subjectivity in this step. I don't know when enough is enough.

Thanks,

adam

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Heng Ye · Teaching Staff · 4 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/qkOdW5PLEeWy0Q7ABZMsnQ/comments/SvYU453gEeW7DQrLk-JR4Q)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

Hi Adam, sorry for late reply.
You are right, the range does influence the threshold point. One way to do it is to adjust the range to [-3.5, 3.5] for both training dataset and test dataset. For the purpose of this class, the answer are set to match those who normalized just once. After all, as long as you get the idea of modeling and threshold, the quiz question serves its purpose.

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(/learn/analytics-excel/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Reply

CG

Christopher Gearin · 16 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/UT4fopSSEeW70BJZtLVfGQ)

Should we be able to see "hints" on the questions after scoring?? I can not!

(/learn/analytics-excel/profiles/9e53f50a38a4f40d9439d9665e4456b3)



Heng Ye · Teaching Staff · 4 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/UT4fopSSEeW70BJZtLVfGQ/comments/FZLpqZ3hEeWJ-hjkaqaKQ)

(/learn/analytics-excel/profiles/31e005485fdaae37d9fb580de1415dc1)

We are aware of that and it is a platform problem. For now, we have included all the hints on the forum. If you have further questions, feel free to drop a line here.

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K

Kshitij · 16 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Fh8rU5S5EeWibw4zhsvrPQ)

Whenever I put data of eegertopia i get the AUC-0.85 but the minimum cost is 1250?what i am doing wrong? and there are many threshold for 1250 and the cost is too high so this is not the correct ans.please help me out

(/learn/analytics-excel/profiles/ad23a918e75ccb758362a75491bc51c8)

SS

Simon Sidorenko · 16 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Fh8rU5S5EeWibw4zhsvrPQ/comments/dZxxzjTAEeWJSqr5zqhl6w)

(/learn/analytics-excel/profiles/7ae1b4ef1c6813fb23f45fd7d6658153)

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Scott Young · 15 days ago (/learn/analytics-excel/discussions/cgaNtlcAEeWL6AoNmIctaQ/replies/Fh8rU5S5EeWibw4zhsvrPQ/comments/-lXu2ZULEeWJSqr5zqhl6w)

(/learn/analytics-excel/profiles/842e7ac9b1e01231933126c4991d6ad)

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